

Initialize the July 2022 heatwave case

Guide 2 of 3 | LIU Zhuo | The Hong Kong University of Science and Technology

Compile both cores using Guide 1 first. This guide prepares geographic fields and initial conditions for a global Hong Kong-refined mesh. Surface updates are optional.

Case reference: [Cao et al. \(2026\), MPAS-Urban and the record-breaking 2022 Hong Kong heatwave](#), ESS Open Archive preprint.

Step 1 | Prepare the mesh and ERA5 inputs

The teaching period is 2022-07-19 00:00 UTC to 2022-07-30 06:00 UTC. Follow the [mesh-generation repository cited by Cao et al.](#) for the global variable-resolution mesh, refined to about 500 m around Hong Kong and coarsening to about 30 km. The inspected case mesh contains 746,053 cells; a newly generated mesh can differ.

Prepare a model-ready spherical MPAS mesh and its matching, unpartitioned graph. Complete the mesh conversion/scaling steps in the mesh workflow; a raw unit-sphere mesh is not the final Earth-sized input. Keep the cell ordering identical between mesh and graph.

Prepare ERA5 [pressure-level](#) and [single-level](#) data as follows:

- Initial conditions: one complete intermediate file, ERA5:2022-07-19_00, containing the required atmospheric, surface and soil fields.
- Optional surface updates: only if you choose Step 7, prepare SST and sea-ice fields every 3 hours from 19 July 00:00 to 30 July 06:00 UTC with prefix SST. Otherwise, the one complete initial-time ERA5 file is sufficient.

This global case does not require lateral boundary conditions.

For WPS preparation, follow the [official WRF/WPS guide](#) and the [MPAS real-data tutorial](#). Use the ERA5-compatible Vtable and inspect the resulting intermediate records. MPAS reads these files directly; WRF met_em files are not used here.

This repository does not distribute a ready mesh or ERA5 archive. After preparation, place the inputs under this layout (the names on the left are the tutorial's local names):

```
$HOME/MPAS/DATA/heatwave2022/  
|-- heatwave2022.grid.nc  
|-- heatwave2022.graph.info  
`-- ERA5:2022-07-19_00
```

Step 2 | Prepare static geographic data

Download the standard geographic archive and the HKUST urban land-cover dataset:

```
export MPAS_ROOT="$HOME/MPAS"  
mkdir -p "$MPAS_ROOT/DATA"  
cd "$MPAS_ROOT/DATA"  
curl -fL -O \  
    https://www2.mmm.ucar.edu/projects/mpas/mpas_static.tar.bz2  
tar -xjf mpas_static.tar.bz2  
cd "$MPAS_ROOT/DATA/mpas_static"  
curl -fL -O \  
    https://github.com/HKUST-MPAS/HKUST-MPAS/releases/download/LCZ_dataset_for_MPAS_Urban/mpas_cg1c_lcz.tar.gz  
tar -xzf mpas_cg1c_lcz.tar.gz
```

The geographic directory must contain the dataset folders, not an extra nested `mpas_static` folder. Check the two indexes below and retain all tile files in the downloaded datasets:

```
test -s "$MPAS_ROOT/DATA/mpas_static/landuse_30s/index"
test -s "$MPAS_ROOT/DATA/mpas_static/mpas_cglc_lcz/index"
```

`landuse_30s` is needed for native-mesh gravity-wave-drag fields, even when the surface land cover is CGLC/LCZ. The reference settings use GMTED2010 terrain and MODIS vegetation fraction, albedo and maximum snow albedo. Land cover comes from the [HKUST LCZ release](#).

Soil data

[Dy and Fung \(2016\)](#) describe an updated WRF soil map built by updating the default map with BNU data. Contact the authors for that dataset and its processing details if it is needed for your experiment. The official MPAS BNU dataset is related, but is not identical to that updated WRF product.

The main workflow here uses the default STATSGO soil-category input supported by HKUST-MPAS v8.2.2. Optional BNU soil data require MPAS v8.3 or later; the HKUST-MPAS version used in this tutorial does not support the `config_soilcat_data` option. Do not add it to this version's namelist.

For a separate setup using MPAS v8.3 or later, download BNU as follows:

```
cd "$MPAS_ROOT/DATA/mpas_static"
curl -fL -O \
  https://www2.mmm.ucar.edu/projects/mpas/bnu_soiltype_top.tar.bz2
tar -xjf bnu_soiltype_top.tar.bz2
test -s bnu_soiltype_top/index
```

In that version only, set `config_soilcat_data = 'BNU'` under `&data_sources` when generating the static file. Changing soil input requires generating a new static file. This optional choice does not change the source version used in the main tutorial.

Step 3 | Create the case directory

Set the paths. Repeat these exports when starting a new shell:

```
export MPAS_ROOT="$HOME/MPAS"
export MPAS_BUILD="$MPAS_ROOT/HKUST-MPAS"
export TUTORIAL="$MPAS_ROOT/MPAS-Urban-Tutorial"
export INPUT="$MPAS_ROOT/DATA/heatwave2022"
export CASE="$MPAS_ROOT/CASES/heatwave2022"
mkdir -p "$CASE"
cd "$CASE"
```

Link the executable, mesh and vertical-level definition:

```
ln -sf "$MPAS_BUILD/init_atmosphere_model" .
ln -sf "$INPUT/heatwave2022.grid.nc" .
ln -sf "$INPUT/heatwave2022.graph.info" .
ln -sf "$MPAS_BUILD/vertical_levels" .
test -s "$INPUT/ERA5:2022-07-19_00"
ln -sf "$INPUT/ERA5:2022-07-19_00" .
```

Step 4 | Partition the mesh

Use METIS `gpmets` on the graph that belongs to your mesh. Here 192 is the number of MPI tasks, matching the reference case; use the resources allocated by your machine.

```
export NPROCS=192
gpmetis heatwave2022.graph.info "$NPROCS"
ls -lh "heatwave2022.graph.info.part.$NPROCS"
```

Optional | Change the number of CPU tasks

For example, to use 112 tasks instead, make a new partition from the unpartitioned graph:

```
export NPROCS=112
gpmetis heatwave2022.graph.info "$NPROCS"
```

Initial-condition generation, optional surface-update generation and the atmosphere run use NPROCS. The static stage below uses one MPI process. Keep the selected NPROCS value in Guide 3. Do not rename `.part.192` to `.part.112`; it contains a different decomposition. The namelist prefix stays `heatwave2022.graph.info.part..`

Step 5 | Generate the static file

```
cp "$TUTORIAL/config/namelist.init_atmosphere" .
cp "$TUTORIAL/config/streams.init_atmosphere" .
nano namelist.init_atmosphere
```

Change `config_geog_data_path` to the full path of your `DATA/mpas_static` directory, ending in `/`. Fortran namelists do not expand `$HOME`. The supplied file marks this one path with `CHANGE_ME`.

These two files start with the static-stage settings. Keep editing the same local files in Steps 6 and 7; do not copy the originals again. In `streams.init_atmosphere`, the input stream reads `heatwave2022.grid.nc` and the output stream writes `heatwave2022.static.nc`.

The static stage has these six switches:

```
&preproc_stages
  config_static_interp = true
  config_native_gwd_static = true
  config_vertical_grid = false
  config_met_interp = false
  config_input_sst = false
  config_frac_seaice = false
/
```

The first two create geographic and native-mesh gravity-wave-drag fields. This stage does not interpolate ERA5 or build the vertical grid.

Run this tutorial's static stage with one MPI process, using the same MPI environment as the build:

```
mpirun -np 1 ./init_atmosphere_model
cp log.init_atmosphere.0000.out log.static.out
ls -lh heatwave2022.static.nc
tail -n 20 log.static.out
```

Expect a non-empty static file and a normal completion in the log. Static interpolation on this large global mesh can take hours, depending on CPUs, storage and data resolution. This is a planning estimate, not a measured duration for this new setup. While waiting, you can explore plotting the mesh using the official MPAS tutorial.

Step 6 | Generate the initial conditions

```
nano namelist.init_atmosphere
nano streams.init_atmosphere
```

Check the start/stop time is 2022-07-19_00:00:00, prefix is ERA5, and vertical levels are 55 with vertical_levels/urban_ZR_75.txt. This file has 56 interfaces, from the surface to 30 km.

The reference uses 38 meteorological levels (37 ERA5 pressure levels plus a surface level) and four soil levels. Set config_nfglevels to match your actual WPS intermediate input. config_use_spechumd = false expects relative humidity; use true only if you prepared specific humidity instead.

```
&preproc_stages
  config_static_interp = false
  config_native_gwd_static = false
  config_vertical_grid = true
  config_met_interp = true
  config_input_sst = false
  config_frac_seaice = true
/
```

This stage reads the existing static file, builds the vertical grid and interpolates ERA5 at the initial time. It does not remake geographic fields.

In streams.init_atmosphere, change the two filename_template values:

- input: heatwave2022.grid.nc to heatwave2022.static.nc.
- output: heatwave2022.static.nc to heatwave2022.init.nc.

Keep the other stream settings unchanged.

```
mpiexec -n "$NPROCS" ./init_atmosphere_model
cp log.init_atmosphere.0000.out log.initial.out
ls -lh heatwave2022.init.nc
tail -n 20 log.initial.out
```

Allow tens of minutes to hours as a broad planning range; actual time depends on the machine and input. Check the log rather than assuming a quiet terminal means the job has stopped.

Step 7 | Optional: generate surface updates

The basic tutorial keeps SST updates off. After Step 6, you can go directly to Guide 3 without preparing any SST: * files or generating heatwave2022.sfc_update.nc.

The original case enabled SST updates. If you choose this option, prepare these additional surface-only intermediate files under \$INPUT, using ERA5 SST and sea-ice fields every 3 hours:

```
SST:2022-07-19_00
SST:2022-07-19_03
...
SST:2022-07-30_06

test -s "$INPUT/SST:2022-07-19_00"
test -s "$INPUT/SST:2022-07-30_06"
ln -sf "$INPUT"/SST:* .
nano namelist.init_atmosphere
```

In `&nhyd_model`, set:

```
config_init_case = 8
config_start_time = '2022-07-19_00:00:00'
config_stop_time = '2022-07-30_06:00:00'
```

In `&data_sources`, retain `config_sfc_prefix = 'SST'` and `config_fg_interval = 10800` (3 hours). Set the six stage switches to:

```
&preproc_stages
config_static_interp = false
config_native_gwd_static = false
config_vertical_grid = false
config_met_interp = false
config_input_sst = true
config_frac_seaice = true
/
```

Keep `streams.init_atmosphere` as it was after Step 6. Its input reads `heatwave2022.static.nc`; the surface stream already writes `heatwave2022.sfc_update.nc` with `output_interval="10800"`. Case 8 uses this surface output, not the initial-condition output.

```
mpiexec -n "$NPROCS" ./init_atmosphere_model
cp log.init_atmosphere.0000.out log.surface.out
ls -lh heatwave2022.sfc_update.nc
ncdump -v xtime heatwave2022.sfc_update.nc
```

Check that `xtime` contains records every 3 hours, covering 19 July 00:00 through 30 July 06:00 UTC. The original atmosphere configuration's hourly surface-stream read interval is retained; it is not a requirement for hourly ERA5 input records.

After completing this optional step, enable SST updates as described in Guide 3. Otherwise, keep the default settings and use the static and initial files generated in Steps 5 and 6.